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SCIENTIFIC CALCULATOR

~ Multi Calculations



 Submitted By :

Jenish Gohel (92300527156)
Ravi Sav (92300527257)

 Internal Guide:

Prof. Tejas Chauhan
Faculty of Computer Applications.

 Institution Address:

Marwadi University,
Rajkot - Morbi Road,
At & PO: Gauridad, Rajkot

INTRODUCTION

In today's digital world, calculators have evolved from simple arithmetic tools to powerful applications capable of handling complex computations. To meet this growing demand, we have developed an **Android-based Scientific Calculator Application** that provides accurate results, an intuitive interface, and multiple calculation utilities in a single app.

The application is designed to serve students, professionals, and general users by offering different types of calculators under one platform. The key calculation modules included in this application are:

- ❑ **Basic Calculator**
Performs basic arithmetic operations such as addition, subtraction, multiplication, and division.
- ❑ **Scientific Calculator**
Supports advanced mathematical operations such as trigonometric functions (sin, cos, tan), logarithmic functions (log, ln), square root, power functions, and degree/radian mode.
- ❑ **Age Calculator**
Calculates the exact age of a user based on the provided date of birth.
- ❑ **Loan Calculator**
Calculates total interest and total payable loan amount based on principal amount, rate of interest, and time period.

The application is developed using Java and XML in Android Studio, following standard Android UI and coding practices. It features a navigation drawer, calculation history, offline functionality, and a clean, modern user interface. The app has been successfully deployed on the Google Play Store and is currently being used by real users.

This project demonstrates the practical application of Android development concepts and delivers a reliable, real-world mobile solution.



PLAY STORE DEPLOYMENT DETAILS

The developed calculator application has been successfully **published on the Google Play Store**, making it publicly accessible to users worldwide. This deployment confirms that the application meets Google's quality, performance, and security standards.

Key Play Store Details:

- **Application Name:** Scientific Calculator
- **Platform:** Android (Google Play Store)
- **Release Status:** Published & Live
- **Availability:** Public (Anyone can download)
- **Distribution Type:** Free Application
- **Play Store Link:** <https://play.google.com/store/apps/details?id=com.jg.scientificcalculator>

Highlights:

- The application is officially uploaded and approved by Google Play.
- It is available for real users and has active downloads.
- All calculator features work **offline**, ensuring reliability.
- Regular updates and improvements can be pushed through the Play Store.
- This deployment transforms the project from an academic application into a **real-world production-level Android app**.

OBJECTIVES OF THE PROJECT



The main objectives of developing the **Scientific Calculator Android Application** are as follows:

- To design and develop a user-friendly calculator application with a clean and intuitive interface.
- To provide basic arithmetic calculations such as addition, subtraction, multiplication, and division.
- To support advanced scientific operations including trigonometric, logarithmic, exponential, and power functions.
- To include multiple calculators such as:
 1. **Basic & Scientific Calculator**
 2. **Age Calculator**
 3. **Loan Calculator**
- To store and manage calculation history for quick reuse of previous results.
- To ensure accurate and fast calculations with support for multiple operands.
- To develop an application that works offline without internet dependency.
- To implement a modern Android UI using Material Design principles.
- To deploy the application on the Google Play Store for real-world usage.



TECHNOLOGIES USED

The following technologies and tools were used in the development of the **Scientific Calculator Android Application**:

- **Programming Language :**
Java used for implementing application logic, calculations, and event handling.
- **Android SDK :**
Provides core APIs for building Android applications.
- **Android Studio :**
Official IDE used for coding, debugging, testing, and deployment.
- **XML :**
Used for designing user interface layouts and defining UI components.
- **SQLite Database :**
Used to store and manage **calculation history** locally on the device.
- **Material Design Components :**
Used for creating a modern, responsive, and visually appealing UI.
- **Navigation Drawer :**
Implemented to provide smooth navigation between multiple calculators.
- **Google Play Console :**
Used for publishing, managing, and updating the application on the Play Store.

KEY FEATURES



- Multi-Calculator Application**

Includes Basic Calculator, Scientific Calculator, Age Calculator, and Loan Calculator in a single app.

- Basic Arithmetic Operations**

Supports addition, subtraction, multiplication, and division.

- Advanced Scientific Operations**

Trigonometric functions: **sin, cos, tan**

Logarithmic functions: **log, ln**

Power, square, cube, and square root calculations.

- Degree & Radian Mode**

Allows users to switch between **DEG** and **RAD** modes for trigonometric calculations.

- Calculation History**

Automatically saves calculations and allows users to reuse previous results.

- User-Friendly Interface**

Clean, modern UI with Material Design components.

- Navigation Drawer Support**

Easy navigation between different calculators using a side menu.

- Offline Functionality**

Works completely offline without internet connection.

- High Accuracy & Fast Performance**

Provides precise results with instant response time.

- Play Store Published Application**

Successfully uploaded on Google Play Store and available for public download.

MODULES IN CALCULATOR



1. Scientific Calculator Module

- Advanced mathematical and trigonometric functions
- Supports long expressions and multiple operands

2. Age Calculator Module

- Calculates age in years, months, and days
- Simple and user-friendly input design

3. Loan Calculator Module

- Calculates:
 - Principal Amount
 - Total Interest
 - Total Payable Amount
- Displays results in an attractive CardView layout

4. History Module

- Stores previous calculations
- Allows users to reuse past results

5. Navigation Drawer Module

- Easy navigation between different calculator modules
- Includes About Us section

TECHNICAL DESCRIPTION



1). Hardware Requirements (User Side) :

- Device Type : Android Smartphone
- Processor : Any standard mobile processor
- RAM : Minimum 2 GB
- Internal Storage : Minimum 20 MB free space
- System Type : 64-bit
- Input Method : Touch Screen
- Display : Any standard smartphone display

2). Software Requirements (User Side) :

- Operating System : Android OS
- Minimum Android Version : Android 6.0 (Marshmallow) or above
- Application Type : Mobile Application
- Installation Source : Google Play Store
- Internet Requirement : Required only for download
- Working Mode : Offline

3) Application Environment :

- Platform : Android
- Category : Utility Application
- User Interface : Touch-based graphical interface
- Database : SQLite (internal, invisible to user)

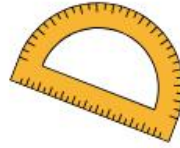
4) Compatibility :

- Supported Devices : Android smartphones and tablets
- Supported Screen Sizes : All standard Android screen sizes
- Supported Orientation : Portrait Mode

APP FEATURE SCREEN DESIGN



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Calculator

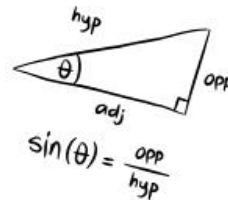
All in one calculator

Scientific Calculator • Age Calculator • Loan Calculator



$$\tan(60^\circ) = \sqrt{3}$$

%



APP LOGO PICTURE



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1. SCIENTIFIC CALCULATOR SCREEN



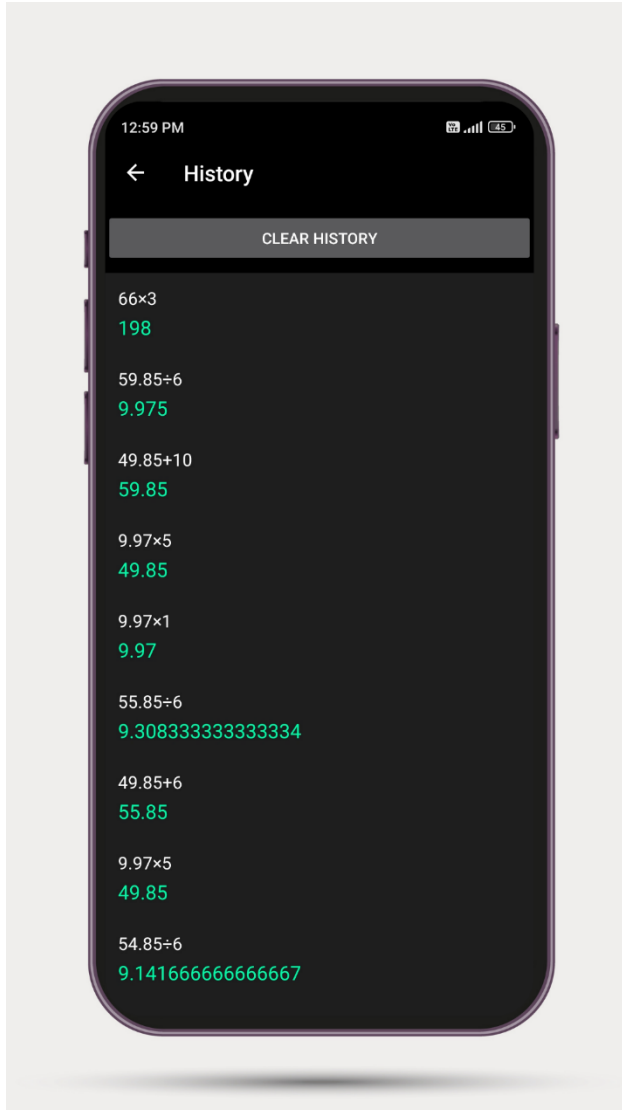
- This screen is main screen that comes when you start application and its actual screen for Basic and Scientific Calculations

2. CALCULATION SCREEN



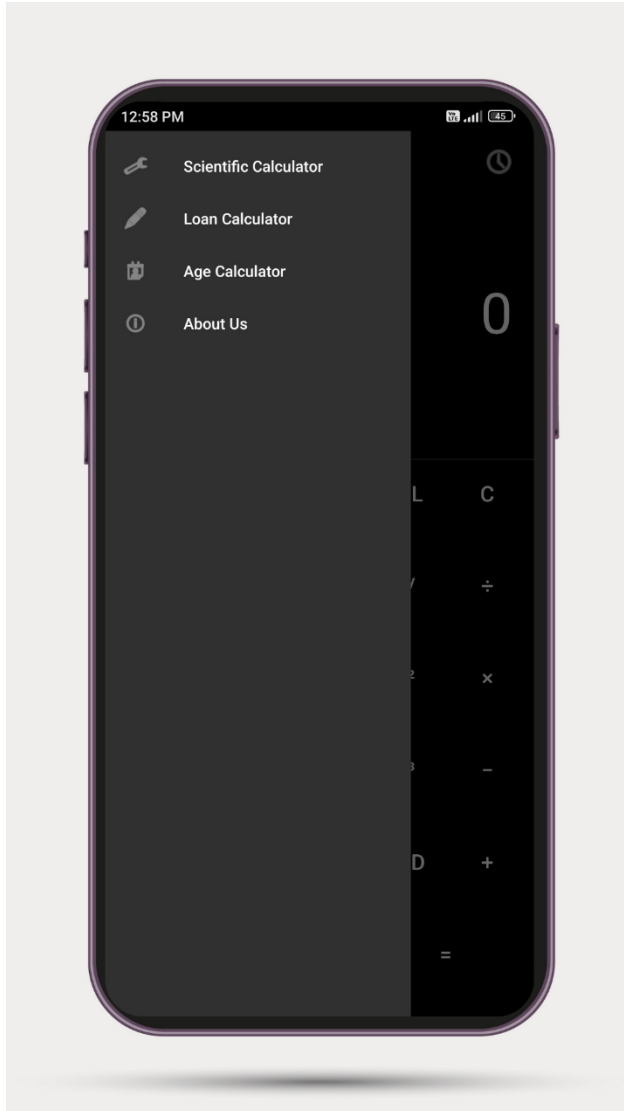
- This screen is main screen it self where all calculations are possible basic to scientific

3. HISTORY SCREEN



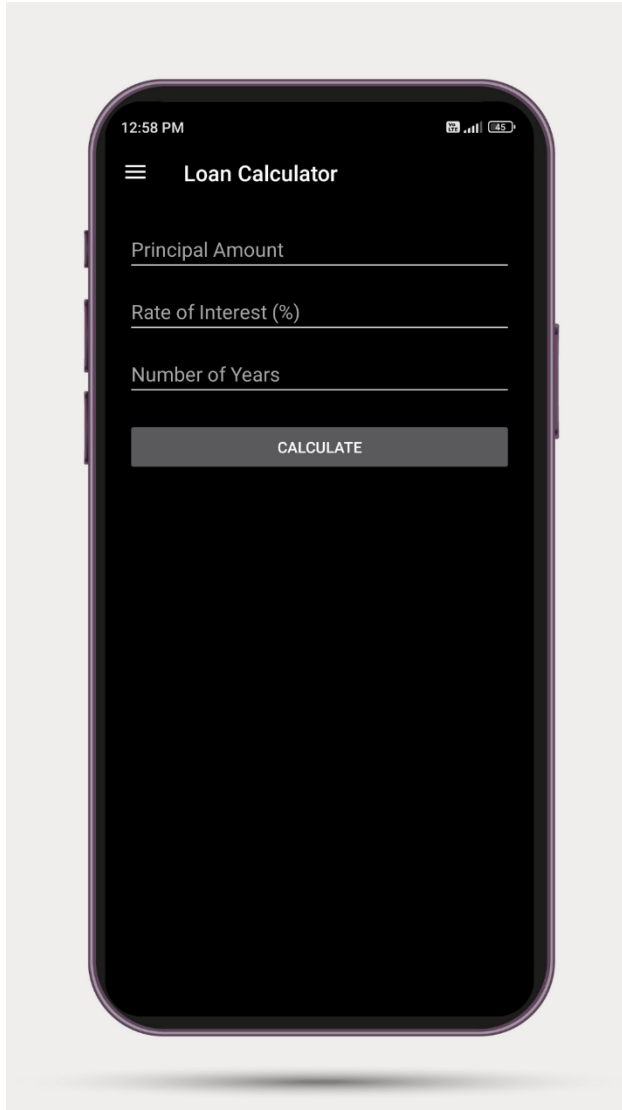
- This screen shows all basic to scientific calculations history with its expression and user can also reuse that expression by clicking on that.

4. DRWAER MENU



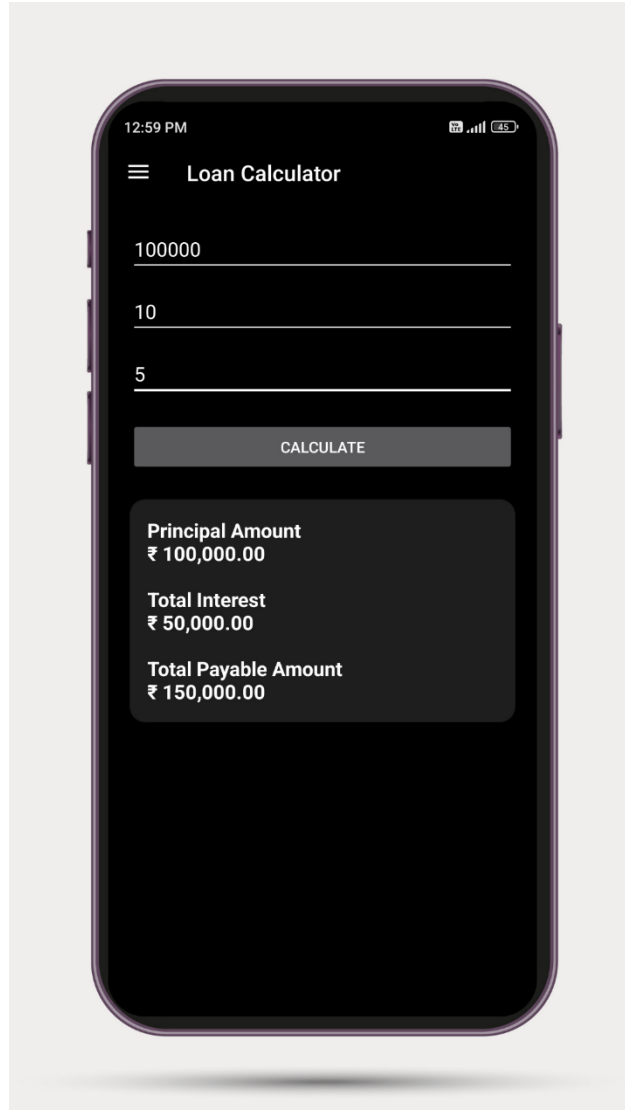
- This is drawer menu of our app that contains 4 options as :
 - Scientific Calculator
 - Loan Calculator
 - Age Calculator
 - About Us

5. LOAN CALCULATOR SCREEN



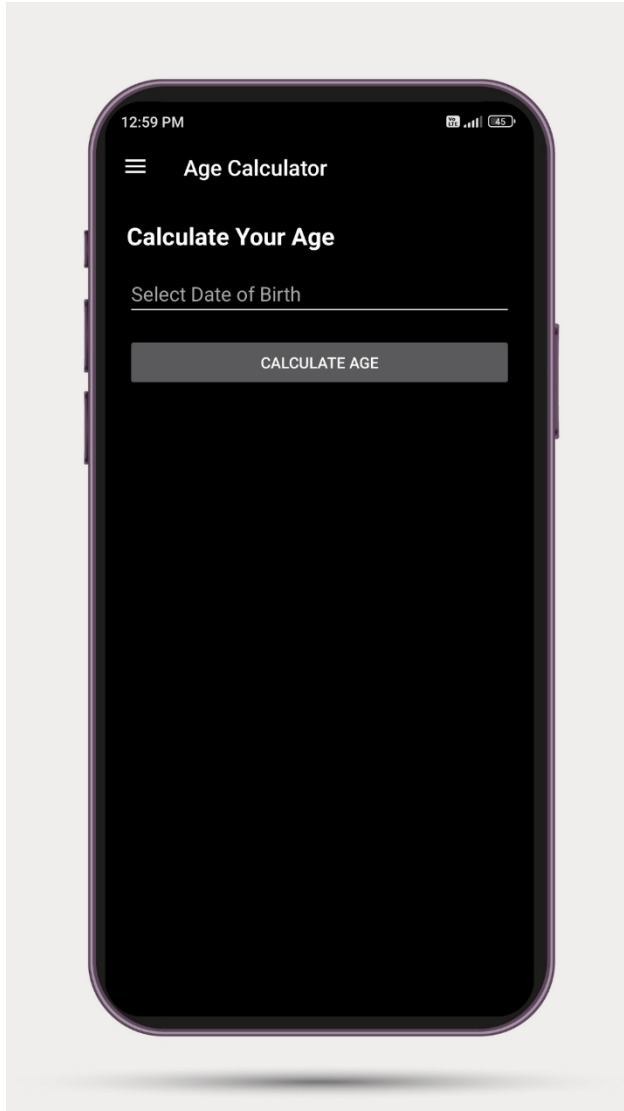
- This screen is Loan Calculator initial screen with 3 input fields for Amount , Rate of Interest and Number of Years with final CALCULATE Button for final result

6. LOAN CALCULATION



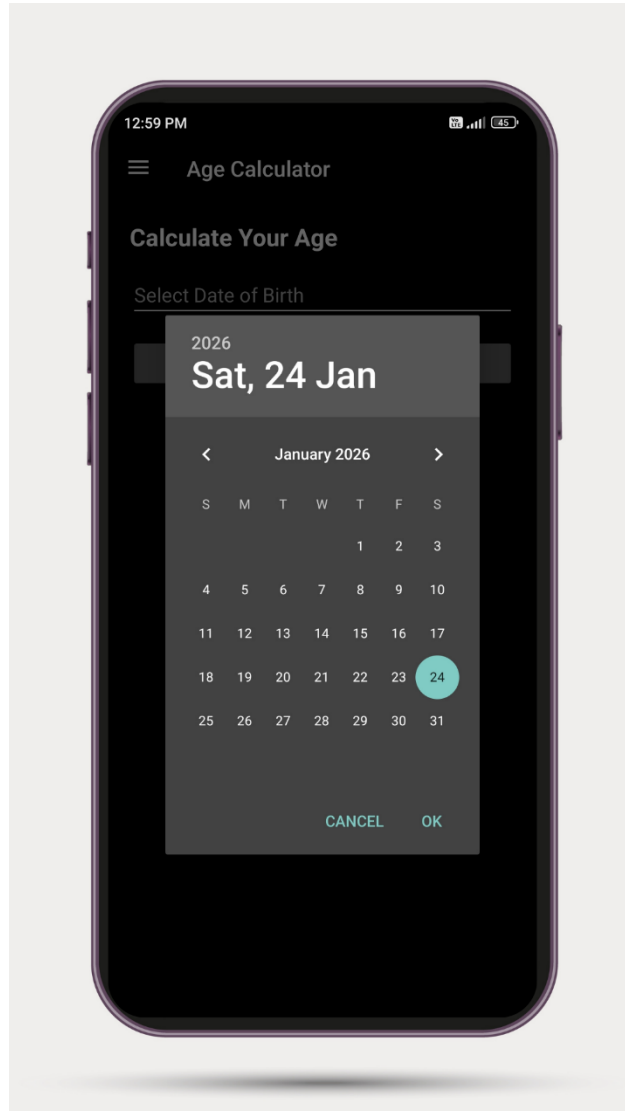
- This screen shows Loan calculation with its given inputs and for that inputs its corresponding outputs

7. AGE CALCULATOR



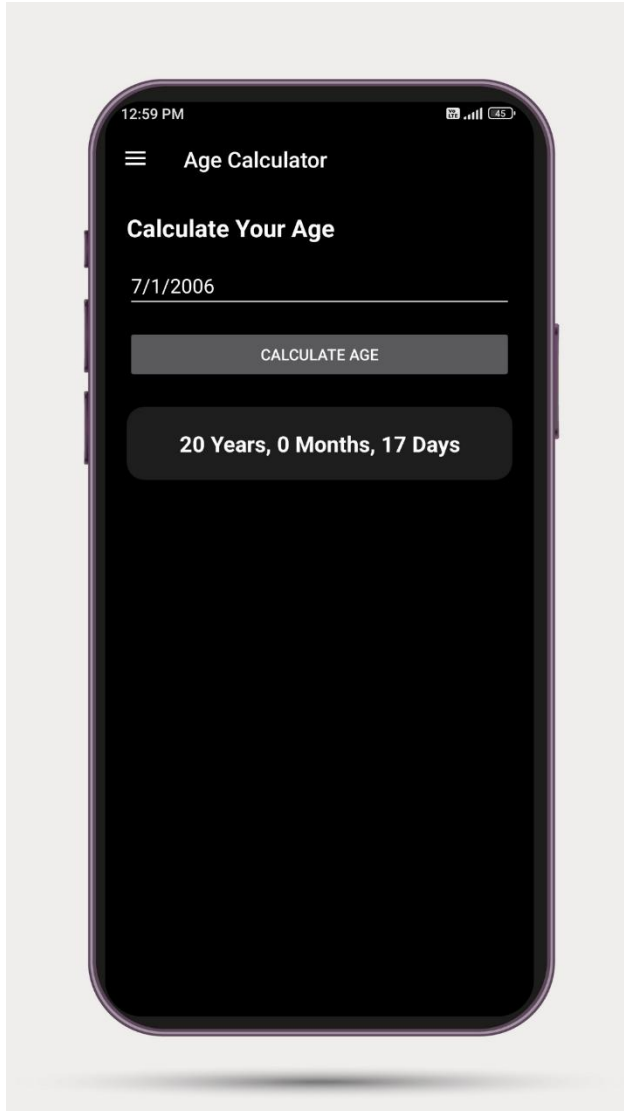
- This screen is Age Calculators initial screen with input field for entering Date of Birth and one CALCULATE AGE button to calculate and display result.

8. DATE OF BIRTH ENTERING



- This screen shows how Date of Birth user can select for calculate His / Her Age.

9. AGE CALCULATION

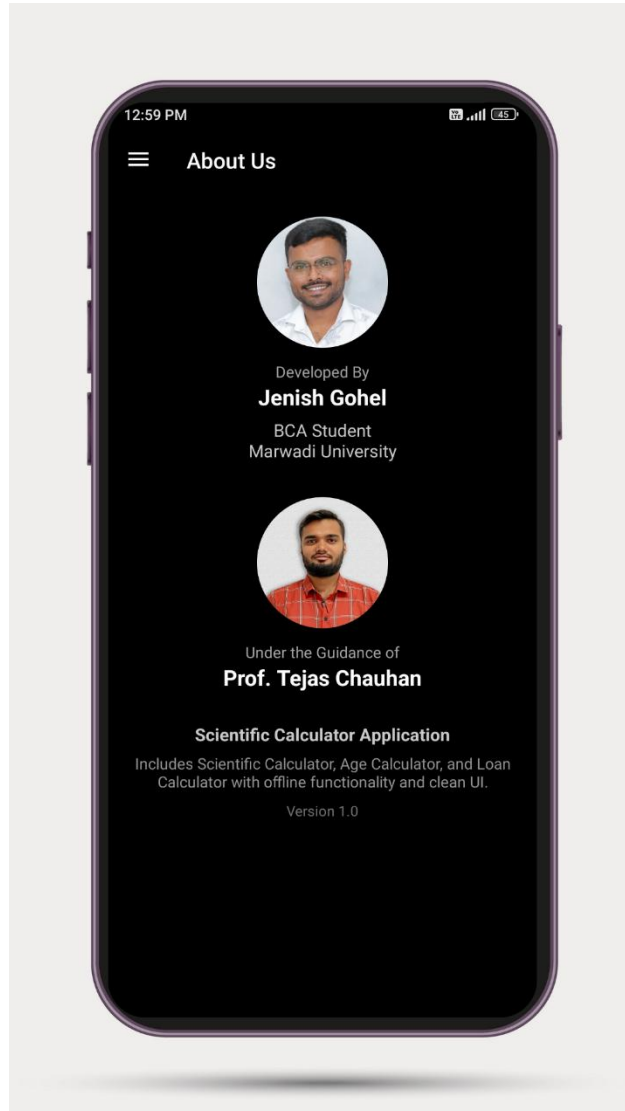


- This screen displays result of user after entering Date of Birth with its Correct Age till today.

10. ABOUT US SCREEN



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- This is about us screen that shows Developer Information along with its Internal Guide Information and Short Application Introduction with Version

DATABASE DESIGN



The Calculator application uses a **local database** to store calculation history.

The database is implemented using **SQLite**, which is lightweight, fast, and suitable for mobile applications.

Database Type :

- Database : **SQLite**
- Storage Location : Internal storage of the Android device
- Mode : Offline
- Access : Application-only (not visible to users)

Purpose of Database :

- To store calculation history
- To allow users to view previous calculations
- To enable reuse of past results
- To improve user convenience and efficiency

Advantages of SQLite in This Project :

- No internet connection required
- Fast data access
- Low storage consumption
- Secure and device-local storage

DATABASE DESIGN



Database Structure :

The application contains one main table for storing history records.

Table Name : **history**

Field Name	Data Type	Description
id	INTEGER	Unique record ID (Primary Key)
expression	TEXT	Calculation entered by the user
result	TEXT	Final computed result
timestamp	TEXT	Date and time of calculation

Primary Key :

- **id** (Auto-incremented)

Data Flow :

- User performs a calculation
- Expression and result are generated
- Data is stored in SQLite database
- User can view history anytime from History screen

LEARNING DURING PROJECT



During the development of the Calculator application, we gained practical knowledge and hands-on experience in various technical and non-technical areas.

1. Android Application Development

- Learned to design Android applications using **Java**
- Understood the use of **Activities, Layouts, and UI components**
- Implemented **multiple screens** using Intent navigation

2. User Interface (UI) Design

- Designed responsive layouts using **XML**
- Learned to manage **screen alignment, scrolling, and overlapping issues**
- Improved user experience with **Material Design components**

3. Mathematical Logic Implementation

- Implemented logic for:
 1. Basic arithmetic calculations
 2. Scientific operations (sin, cos, log, power, etc.)
 3. Loan and age calculations
- Handled edge cases like invalid inputs and operator errors

4. Database Handling

- Learned to use **SQLite database** in Android
- Stored and retrieved **calculation history**
- Implemented CRUD operations using a database helper class

LEARNING DURING PROJECT



5. Debugging and Problem Solving

- Fixed runtime errors and logical bugs
- Resolved issues such as:
 - Auto-scroll behavior
 - Scientific notation formatting
 - Navigation drawer problems
- Improved accuracy and stability of the application

6. Navigation & App Flow

- Implemented **Navigation Drawer**
- Learned smooth switching between multiple calculator modules
- Maintained consistent UI across different screens

7. App Deployment Knowledge

- Learned steps to:
 1. Generate signed APK / AAB
 2. Upload application to **Google Play Store**
 3. Manage app versions and updates

8. Teamwork & Project Management

- Improved planning and time management skills
- Learned to document code and features clearly
- Gained confidence in explaining technical concepts during presentations

Overall Learning Outcome

This project enhanced our **technical skills, logical thinking, and problem-solving ability**, and gave us real-world experience in developing and publishing a complete Android application.

BIBLIOGRAPHY



Online References :

- [1] Android Developers, *Official Android Developer Documentation*.
[Online]. Available: <https://developer.android.com/>
[Accessed: Sept. 10, 2026].
- [2] Oracle Corporation, *Java SE Documentation*.
[Online]. Available: <https://docs.oracle.com/javase/>
[Accessed: Sept. 10, 2026].
- [3] Google Developers, *Material Design Guidelines*.
[Online]. Available: <https://material.io/design>
[Accessed: Sept. 10, 2026].
- [4] SQLite Consortium, *SQLite Official Documentation*.
[Online]. Available: <https://www.sqlite.org/docs.html>
[Accessed: Sept. 10, 2026].
- [5] W3Schools, *Java and Android Programming Tutorials*.
[Online]. Available: <https://www.w3schools.com/>
[Accessed: Sept. 10, 2026].
- [6] YouTube, *Educational Android Development Tutorials*.
[Online]. Available: <https://www.youtube.com/>
[Accessed: Sept. 10, 2026].
- [7] Stack Overflow, *Community-Based Troubleshooting Platform for Android Developers*.
[Online]. Available: <https://stackoverflow.com/>
[Accessed: Sept. 10, 2026].
- [8] Google Play Console, *Android App Publishing Guidelines*.
[Online]. Available: <https://play.google.com/console/>
[Accessed: Sept. 10, 2026].

BIBLIOGRAPHY

Offline References :

- [9] Study materials and lecture notes provided by the **Faculty of Computer Applications**, Marwadi University, India.
- [10] Practical learning and hands-on experience gained during **Android application development laboratory sessions**.
- [11] Continuous guidance, reviews, and technical suggestions provided by the **project guide and faculty members** during project development.
- [12] Concepts and principles learned from **Mobile Application Development, Java Programming, and Software Engineering** subjects as part of the academic curriculum.

CONCLUSION



The **Scientific Calculator Android Application** successfully delivers a modern, accurate, and user-friendly mobile calculator designed using **Android Studio with Java and XML**. By integrating multiple calculator modules into a single application and enhancing it with a clean UI and smooth navigation, the system provides an efficient digital solution for everyday and advanced mathematical calculations.

Key Outcomes:

- Successfully developed a Scientific Calculator Android application
- Implemented basic arithmetic operations such as addition, subtraction, multiplication, and division
- Integrated advanced scientific functions including trigonometric, logarithmic, power, and root operations
- Developed an Age Calculator module for accurate age computation
- Implemented a Loan Calculator module to calculate interest and total payable amount
- Integrated calculation history using SQLite database for result tracking
- Implemented Degree and Radian mode switching for scientific calculations
- Designed a responsive, clean, and user-friendly interface
- Enabled offline functionality without internet dependency
- Successfully published and deployed the application on Google Play Store

Overall, the **Scientific Calculator project** meets its objectives by offering a reliable, feature-rich, and scalable mobile application. The project provides practical exposure to **Android application development, UI/UX design, Java programming, XML layouts, database integration, and real-world app deployment**, making it highly valuable for academic learning and future enhancements.

THANK YOU

